

Displacement Measuring Interferometer

1010623

The IDS3010 is an ultra-precise laser interferometer offering sub-nanometer precision and certified accuracy. Its multi-axis, fiber-based design provides maximum flexibility, enabling the use of different sensor heads and reliable operation even in extreme environments. The system also delivers an outstanding 10-year lifetime, achieved through the combination of a DFB laser and an acetylene reference gas cell, effectively preventing long-term degradation.



Technical Specifications

Sensor	
type	fiber-based laser interferometric measurement system
sensor head environment compatibility	ambient, ultra-high vacuum (UHV), cryogenic
number of sensor axes	3
working distance	0 – 5 m (30 m with optional long-range feature)
target reflectivity	4 - 100 %
max. target velocity	2 m/s
max. measurement bandwidth	10 MHz
sensor resolution	1 pm
noise	see table page 2 ¹
wavelength accuracy (10 years)	10 ppb = 10 nm/m (k = 1) ²
environmental compensation accuracy	200 ppb (k = 1) ²
periodic non-linearities (p-p)	guaranteed 5 nm ³ , typical 3 nm ³
laser frequency stability (60 hours)	10 ppb = 10 nm/m (k = 1) ²
laser class	1
expected lifetime	10 years ²
MTBF (electronics)	51 years ²
warm-up time	not required
absolute position referencing	µm-scale (highly depending on setup stability)
Interfaces	
analog interfaces	sin/cos, linear analog (optional)
digital interfaces	AquadB, HSSL, BiSS-C (optional)
real-time interface bandwidth	up to 25 MHz
resolution sin/cos (inc.)	freely assignable; 1 pm - 2 ²⁴ pm
resolution AquadB (inc.)	freely assignable
resolution HSSL (abs.)	8 - 48 bit
Software and Communication Interfaces	
web browser	no software drivers necessary as all functionality is accessible via Ethernet
APIs	C, C#, LabVIEW, Python, MATLAB, JSON-RPC
WAVE-Software (optional)	data streaming and measurement visualization (up to 1 MHz)
Controller Hardware	
chassis	55 x 52 x 195 mm ³
weight	730 g
power supply	12 VDC
power consumption	8 W
laser source (measurement laser)	DFB laser
laser output power (measurement laser)	max. 400 µW
laser wavelength (measurement laser)	1 530 371 078 fm
laser source (alignment laser)	laser diode
laser output power (alignment laser)	< 0,5 mW
laser wavelength (alignment laser)	650 nm
wavelength reference	acetylene gas absorption cell
controller operating environment	ambient conditions

¹⁾ expected values based on attocube internal measurements with limited statistic certainty

²⁾ expected values based on component specifications and/or certified measurements by official standards and metrology institutes with limited statistic certainty

³⁾ with sensor head C7.6 and stable alignment at 10-1000 mm working distance

Precision

Precision is typically expressed by the standard deviation σ of repeated measurements and is determined by noise, repeatability, and long-term stability.

Noise

Noise is quantified by the standard deviation σ of random signal variations within a defined bandwidth during static or dynamic measurements. Besides intrinsic electrical noise of the IDS (Table 1), external influences such as vibrations and environmental changes contribute to the overall noise level (Figure 1). These external contributions are mainly observed below 1 kHz. Furthermore, the noise level increases with larger measurement distances (Figure 2). For even more detailed information please read our [TechNote MAS-16 IDS-Noise-Analysis](#).

Configuration	Bandwidth	RMS noise (k=1)
Invar cavity, 40 mm	1 kHz – 10 kHz	0.033 nm
Invar cavity, 40 mm	10 kHz – 200 kHz	0.104 nm
free-space, 200 mm	1 kHz – 10 kHz	0.243 nm
free-space, 1000 mm	1 kHz – 10 kHz	0.940 nm

Table 1: Achievable noise levels for different measurement setups as a function of measurement distance under ambient conditions. Evaluation of frequencies above 1 kHz primarily represents the intrinsic noise contribution of the IDS.

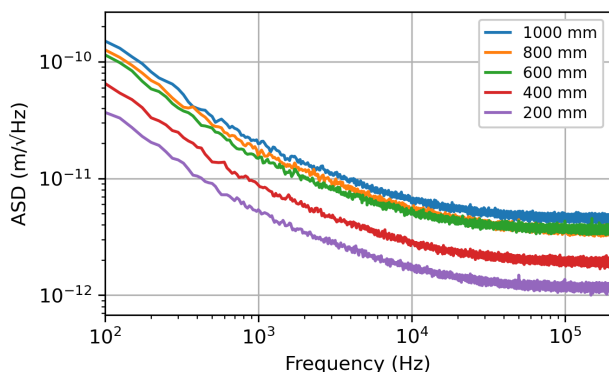


Figure 1: The measurements were conducted in typical laboratory environment. The displayed noise performance is influenced by external noise sources like the mechanical stability of the setup and environmental conditions. The chart therefore displays a realistically achievable performance in comparably simple setups and does not show the limitations of the IDS itself.

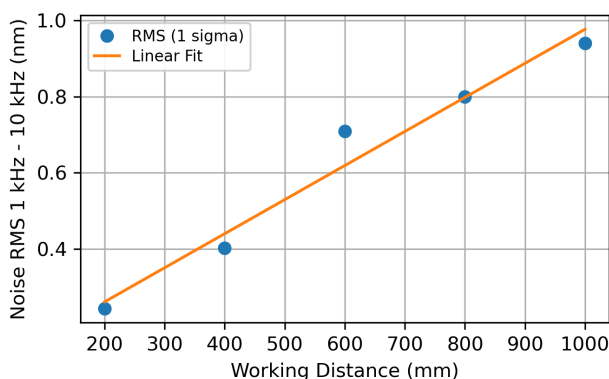


Figure 2: The measurements were also conducted in typical laboratory environment. The data demonstrate an increase of the noise level with rising distance including the contribution of external noise sources. The graph shows the 1 sigma RMS noise values in the 1 kHz - 10 kHz range together with a corresponding linear fit.

Laser frequency stability

Ensuring the reference wavelength provided by the acetylene gas cell is kept with highest stability is crucial for the precision of interferometric measurements, as any wavelength changes impact the displacement value. The IDS offers an outstanding PTB (German National Metrology Institute) certified laser frequency stability of less than 10 ppb (0 – 60 hours; k=1).

Accuracy

Accuracy describes the closeness of a measured displacement value to the true (traceable) displacement of the target. In practice, accuracy is typically expressed in terms of measurement uncertainty over the specified measurement range. In stable vacuum conditions, the measurement uncertainty is primarily limited by laser frequency (vacuum wavelength) stability and residual interferometric non-linearities such as periodic errors. In standard ambient conditions, the dominant contribution to the measurement uncertainty is the uncertainty of the refractive index compensation of air. For applications with significant temperature variations thermal drift of the interferometer unit must also be considered. For even more detailed information please read our [TechNote_MAS-21 IDS interferometric system measurement Uncertainty](#).

Vacuum wavelength accuracy

This parameter describes the accuracy of the acetylene-based reference wavelength at the selected absorption line, which serves as the primary wavelength reference for the interferometric measurement. Any offset in the reference wavelength results in a proportional scale factor error in the displacement measurement. The wavelength accuracy of the gas cell used in the IDS system has been certified by PTB (German national metrology institute): Vacuum wavelength accuracy = 10 ppb (k=1) The specified wavelength accuracy is valid over a service lifetime of up to 10 years.

Periodic Nonlinearities

Periodic nonlinearities are inherent to optical interferometric displacement measurements and their impact on measurement performance primarily depends on the implemented mitigation and compensation strategies. The IDS features a highly adaptive Automatic System Compensation that corrects nonlinearities caused by beam non-orthogonality (cosine error) as well as multiple reflections between the target and the fiber end face. For shorter measurement distances, selecting the Optimized Linearity Mode further reduces nonlinearities by optimizing the signal generation scheme to minimize these effects at their source (resulting in a maximum travel speed of 0.9 m/s). In general, the relative impact of periodic nonlinearities decreases with increasing measurement distance (see Table 2). In addition, the IDS offers a Non-Linearity Compensation Feature, which allows to analyze the non-linearities and to generate a LUT (look up table) for static correction. Depending on the sensor head the validity of the LUT only applies for a reduced travel range (see Table 3).

Sensor Head	Working distance	Typical Non-Linearity
C7.6	1000 mm	2 nm (p-p)
C1.6	300 mm	5 nm (p-p)
F40	48 mm	5 nm (p-p)

Table 2: Typical Non-Linearity at exemplary working distances using the Optimized Linearity Mode for C7.6 sensor head measured on retroreflector, and C1.6 as well as F40 sensor heads measured on a mirror.

Sensor Head	Working distance	Typical Non-Linearity after compensation
C7.6	1000 mm	1 nm (p-p)
C1.6	300 mm	2 nm (p-p)
F40	48 mm	3 nm (p-p)

Table 3: Similar measurement as in Table 2, but with activated LUT compensation (optional firmware feature).

Ambient accuracy

In interferometric measurements in air, the system measures optical path length rather than purely geometric displacement. This depends on the refractive index of air (n), which is influenced by temperature, pressure, humidity, and air composition. Since the refractive index applies along the entire optical beam path, environmental variations directly affect the measured absolute distance and displacement result. An Environmental Compensation Unit (ECU) compensates for this effect by measuring temperature (T), pressure (P), and humidity (H), calculating the refractive index of air, and applying real-time corrections to the interferometer output. The achievable ambient measurement performance is therefore primarily limited by the overall ECU uncertainty contribution, which is determined by the accuracy of the individual ECU sensors (T , P , H) as well as the uncertainties introduced by the calibration chain, resulting in the specified environmental compensation accuracy. For further information on individual TPH sensor accuracy see dedicated [ECU specification sheet](#).

Calibration Method	Accuracy
Factory release (standard)	200 ppb ($k=1$)
Accredited laboratory (optional)	100 ppb ($k=1$)

Table 4: Achievable ambient accuracy by refractive index compensation based on different calibration methods.

Thermal uncertainty IDS unit

Unlike the optical measurement path, which is typically well controlled, the interferometer control unit is often exposed to a less stable thermal environment, which may affect laser wavelength stability and overall system performance. However, the IDS design is optimized for high thermal stability. Based on thermal cycling tests in a climate chamber, no measurable temperature dependency of the measurement result was observed in the range of 17–27 °C.

Lifetime & recalibration

Compared to HeNe laser-based interferometers, the IDS does not require laser tubes, which are subject to aging, require maintenance, and periodic replacement in continuous 24/7 operation. The IDS uses a telecommunication-wavelength DFB laser in combination with an acetylene reference gas cell (specified lifetime of 10 years). This enables stable system performance without the need for laser recalibration over a 10-year period. For applications in ambient environments, periodic recalibration of the Environmental Compensation Unit (ECU) is required within a 1-year cycle to maintain the specified ambient measurement performance. The electronics of the IDS achieve a Mean Time Between Failure (MTBF) of 51 years according to the Siemens SN 29500 standard, highlighting the long-term reliability of the system.

Master slave dependency

The working distances depend on the combination of selected measurement axes. Depending on the working distance of the master axis, the working distances of the remaining axes are restricted to the range shown in figure 3.

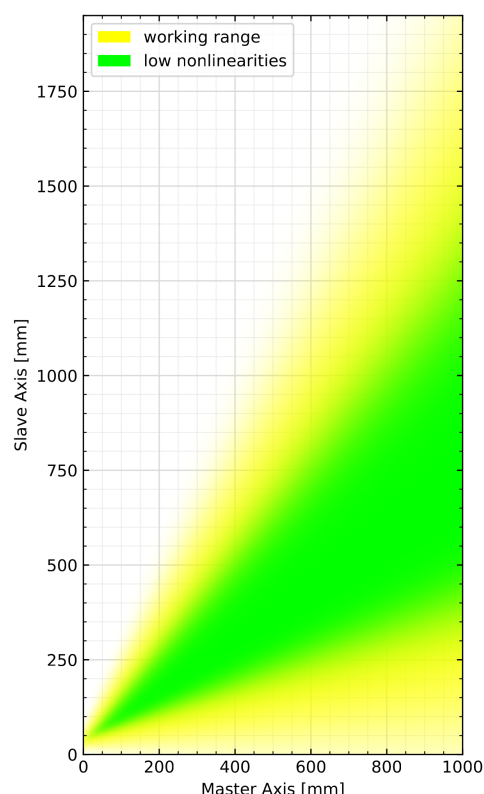


Figure 3: Master slave dependency and impact on the non-linearity performance.

